

Discrete distributions

Match the story to a named PMF, then read off its mean and variance.

THE QUESTION

Which PMF fits a yes/no outcome, a count of successes, a wait or a count of events, and what are its mean and variance?

THE INTUITION

A few stories cover most discrete data: a yes/no trial, a pick among K labels, successes in n tries, the wait for a success, events at a steady rate. Each story fixes the PMF up to one or two parameters.

KEY TAKEAWAY

Name the story first. Bernoulli: mean p . Binomial: np and $np(1-p)$. Geometric: $1/p$ tries. Poisson: mean = variance = λ .

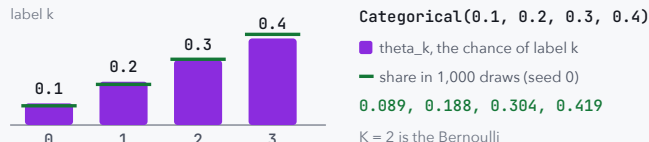
K labels: categorical → K = 2: Bernoulli(p) → n tries: binomial → wait for a success: geometric → $n \rightarrow \infty, np = \lambda$: Poisson

1 One yes/no trial: Bernoulli(p)



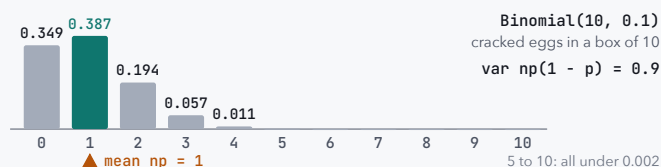
$X = 1$ (a success) with chance p , else 0: $p_X(x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$. Mean p , variance $p(1-p)$: largest at $p = 1/2$, and 0 at $p = 0$ or 1, where nothing is uncertain. Coins, pass/fail checks, clicks. The indicator of any event A is $\text{Bernoulli}(P(A))$. To simulate, `rng.random(n) < p` gives n draws; their mean estimates p .

2 One pick among K labels: categorical



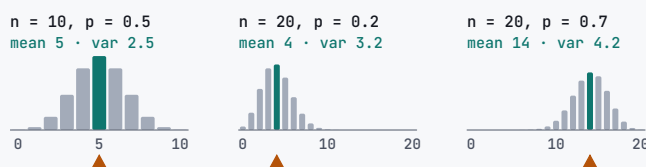
$P(X=k) = \theta_k$ with each $\theta_k \geq 0$ and $\sum_k \theta_k = 1$; a fair die has $\theta_k = 1/6$. Labels such as "cat" and "dog" have no order, so a mean needs numeric labels (the die: mean $7/2$, variance $35/12$). Code numbers the labels 0 to $K-1$: here `torch.distributions.Categorical` has support 0 to 3 and rejects 4.

3 Successes in n tries: Binomial(n, p)



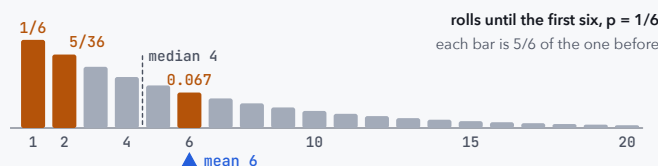
$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$, $k = 0, \dots, n$: there are $\binom{n}{k}$ orders of k successes, each with chance $p^k(1-p)^{n-k}$. It needs a fixed n , independent tries and one p . Ten eggs, each cracked with chance 0.1: $P(0) = 0.9^{10} = 0.349$, $P(1) = 10 \cdot 0.1 \cdot 0.9^9 = 0.387$, $P(2) = 45 \cdot 0.01 \cdot 0.9^8 = 0.194$; 3 or more, 0.070.

4 A binomial is a sum of n Bernoullis



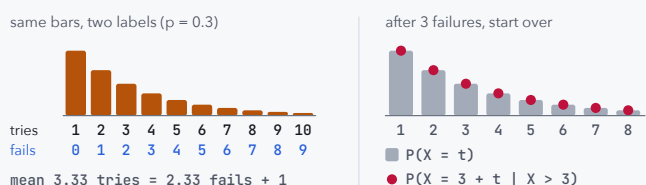
$X = X_1 + \dots + X_n$ with independent $X_i \sim \text{Bernoulli}(p)$, so the mean is np and the variance $np(1-p)$, n times the Bernoulli's (proof on the expectation sheet). The chart is symmetric at $p = 1/2$; for small p its tail runs right, for large p left. The mode (teal) sits next to the mean (triangle). 100 items with 5% defects: mean 5, variance 4.75.

5 Tries until the first success: Geometric(p)



$P(X=k) = (1-p)^{k-1}p$, $k = 1, 2, \dots$: $k-1$ failures, then a success. Rolling for a six: $P(1) = 1/6$, $P(2) = 5/36$, $P(6) = 0.067$; the mode is always 1. Mean $1/p = 6$ rolls, variance $(1-p)/p^2 = 30$. As $P(X > k) = (1-p)^k$, a six comes by roll 4 more often than not (0.518), yet 6 rolls miss it with chance 0.335.

6 Count tries or failures; either way it forgets

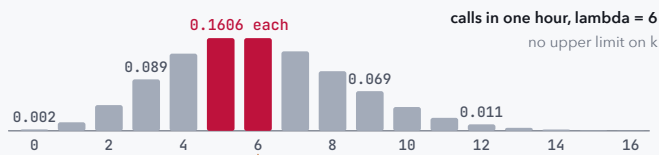


$Y = X - 1$ counts the failures: $P(Y=k) = (1-p)^k p$, $k = 0, 1, \dots$, mean $(1-p)/p$, same variance. NumPy's `rng.geometric` counts tries (smallest value 1); `torch.distributions.Geometric` counts failures (it returns 0s). Memoryless: $P(X > s+t | X > s) = (1-p)^t = P(X > t)$, so after 3 misses the wait starts over. In 100,000 simulated waits with $p = 0.3$: $P(X > 5 | X > 3) = 0.489$; exact $0.7^2 = 0.49$.

Poisson counts, the binomial limit, and which to use

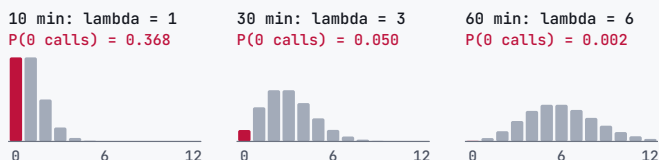
Events at a steady rate give a Poisson count. It is the binomial with many tries and a small chance per try.

7 Counts at a steady rate: $\text{Poisson}(\lambda)$



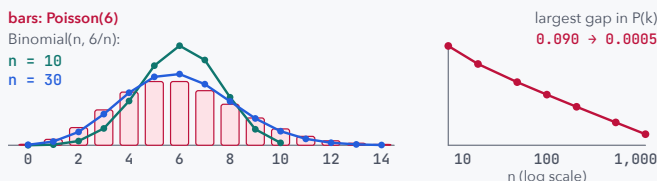
$P(X = k) = \lambda^k e^{-\lambda} / k!$, $k = 0, 1, 2, \dots$, with λ the average count per window. A help desk gets 6 calls an hour: $P(5) = P(6) = 0.1606$, since $P(k)/P(k-1) = \lambda/k$ is 1 at $k = 6$. Mean = variance = λ . Zero calls has chance $e^{-6} = 0.0025$; 10 or more, 0.084. It assumes events come one at a time, independently, at a constant rate.

8 λ grows with the window



6 an hour is 3 per half hour and 1 per 10 minutes, so $P(\text{no call in 10 min}) = e^{-1} = 0.368$. Counts in separate windows are independent and add: $\text{Poisson}(a) + \text{Poisson}(b) = \text{Poisson}(a + b)$. A check on count data: a sample variance far above the sample mean (clustered events) means a Poisson fits poorly. In PyTorch, $\text{Poisson}(6.).\text{log_prob}(k).\text{exp}()$ gives the chart's values for a tensor k .

9 Poisson is the limit of $\text{Binomial}(n, \lambda/n)$



Cut the hour into n slots, each holding a call with chance λ/n :

$$\binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} = \frac{n(n-1)\dots(n-k+1)}{n^k} \cdot \frac{\lambda^k}{k!} \cdot \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-k}$$

As $n \rightarrow \infty$ the first and last factors go to 1 and $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$. For $\lambda = 6$, $P(6)$ is 0.251 at $n = 10$ and 0.161 at 1,000; for large n the largest gap is about $0.5/n$. For large n and small p use $\lambda = np$: 2,000 characters, each mistyped with chance 0.001, give $P(0) = 0.13520$ against $e^{-2} = 0.13534$.

10 PMFs in code, checked by sampling

```
from math import comb, exp, factorial
comb(10, 1) * 0.1 * 0.9**9 # 0.3874
6**6 * exp(-6) / factorial(6) # 0.1606
rng = np.random.default_rng(0)
x = rng.poisson(6, 100_000)
x.mean(), x.var() # 6.002, 6.034
g = rng.geometric(1/6, 100_000)
g.min(), g.mean() # 1, 5.994: tries
```

11 Which distribution? Match the story

Story	Model	Values	$P(X = k)$	Mean	Variance
one yes/no trial	Bernoulli(p)	0, 1	$p^k(1-p)^{1-k}$	p	$p(1-p)$
one pick among K labels	Categorical(θ)	K labels	θ_k	.	.
successes in n tries	Binomial(n, p)	0, ..., n	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
tries until the first success	Geometric(p)	1, 2, ...	$(1-p)^{k-1} p$	$1/p$	$(1-p)/p^2$
tries until the r th success	neg. binomial(r, p)	$r, r+1, \dots$	$\binom{k-1}{r-1} p^r (1-p)^{k-r}$	r/p	$r(1-p)/p^2$
events in a window at rate λ	Poisson(λ)	0, 1, 2, ...	$\lambda^k e^{-\lambda} / k!$	λ	λ

12 Pause and think

Guess all 5 true/false questions. P (all right)? Binomial(5, 1/2) at 5: 1/32. Mean 2.5 right.
Aces among 5 cards dealt from a deck: binomial? No: without replacement the chance changes each card.

Counts have mean 6 and variance 20. Poisson? A poor fit: a Poisson has variance equal to its mean.
A Bernoulli has variance 0.21. What is p ? 0.3 or 0.7: $p(1-p)$ is the same for both.

No six in 10 rolls. Expected extra rolls? Still 6: the geometric forgets the misses.
2 sales a day on average (Poisson). $P(\text{none in two days})$? $\lambda = 4$: $e^{-4} = 0.018$.

Means: Bernoulli p · binomial np · geometric $1/p$ · Poisson λ , also its variance · Binomial($n, \lambda/n$) $\rightarrow$ Poisson(λ)

Source: the course notebook "PMF and some common discrete distributions" (Bernoulli, categorical with $\theta = (0.1, 0.2, 0.3, 0.4)$, binomial with 10 items and $p = 0.1$, Poisson with rate 6, geometric with $p = 0.3$) and the slides "Discrete probability distributions" (means, variances, uses), nipunbatra.github.io/psdv-teaching. The other examples and all numbers are our own, computed by the sheet's figure script; simulations use seed 0. Next: the CDFs and sampling sheet.